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Control of chaos in Lorenz system

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Abstract

A strategy is proposed to control Lorenz chaos, whereby a sliding surface is assigned such that a sliding mode motion occurs when the proposed control law is applied. The results show that only part of the system state can be regulated arbitrarily, due to the structure of the Lorenz system and the applied control variable. The paper also provides a further discussion on the resulting system behavior, which is not addressed in the literature.

1. Introduction

Chaos has been found in many engineering systems. A fundamental characteristic of a chaotic system is its extreme sensitivity to initial conditions; that is, small differences in the initial state can lead to extraordinary differences in the system state. Chaotic behavior is irregular, complex and, in mechanical systems, generally undesirable. In many practical mechanical applications, improved system performance or the avoidance of fatigue failure requires controlling the system so that chaos is removed, giving the system stable and predictable behavior. Therefore, within the research area of non-linear dynamics, the controlling or ordering of chaos is receiving increasing attention. In the field of chaos analysis and control, the Lorenz system is considered a paradigm, since it captures many of the features of chaotic dynamics. It is perhaps the simplest model for the dynamics of convective layers and the dynamics of closed convection loops. The Lorenz equation models an incompressible fluid between two parallel horizontal boundaries, with the lower boundary at a higher temperature than the upper boundary.

Chaos control has been of broad interest since the early 1990s, the most popular case being Lorenz chaos control. In 1990, Ott et al. [1] showed that a chaotic attractor can be converted to any one of a large number of possible attracting time-periodic motions by making only small time-dependent perturbations of an available system parameter. Luce and Kernevez [2] studied the controllability of Lorenz chaos using the optimal control approach. In their work, an approximate solution was provided for bringing a state to another state. At the same time, Vincent and Yu [3] used a bang–bang controller to control Lorenz chaos, showing that it is possible to drive a Lorenz chaotic system to one of the unstable equilibrium points. Hartley and Mossayebi [4] presented a classical analysis of controlling Lorenz equations, with their analysis based on perturbation linearization. In 1994, Gallegos [5] considered the input state and input–output feedback linearization of Lorenz equations, using the non-linear geometric control theory to control Lorenz chaos. Russell [6] studied the non-linear control of Lorenz chaos. A heuristic approach was adopted to obtain a switching control law to stabilize Lorenz chaos. Yeap and Ahmed [7] provided a sub-optimal feedback controller implemented by a multilayer feed-forward neural network to control Lorenz chaos.

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Recently, Fuh and Tung [8] presented an effective approach for controlling Lorenz chaos by using a differential geometric method. This proposed method could control Lorenz chaotic motion not only to a steady state but also to any desired periodic orbit. Yu [9] introduced a control strategy that utilized the switching-in of a two-value Lorenz system parameter. Such switching generates a sliding region such that, once the system state enters the region, the system state will be attracted to an equilibrium point. In 1997, Zeng and Singh [10] used an adaptive control law to drive the state of a Lorenz system to a specified point in the state space.

In most publications regarding Lorenz chaos control, it is considered essential to know the Lorenz model parameters for the successful derivation of a controller. Further, it is often assumed that the system is without external disturbance. If external disturbance is considered, then, in general, only match disturbance is considered. In a practical situation, the parameters of the Lorenz chaotic system may not be known, may be varying in time and may be undergoing either match or mismatch disturbance. Thus, the derivation of a chaos controller for a Lorenz system in the presence of parameter uncertainty and disturbance is an important problem.

This paper considers a Lorenz system with parameter uncertainty and disturbance. The method of sliding mode control is applied to control the Lorenz chaos. Based on this proposed method, a switching surface is designed for Lorenz chaos. The approach provides immunity to parametric uncertainties, without chattering. The results show that the proposed controller regulates the chaotic behavior and accomplishes set point control in spite of uncertain system parameters and external disturbance. The paper also provides a further discussion on the resulting system behavior. It reveals that only part of the system state can be regulated arbitrarily, due to the structure of the Lorenz system and the applied control variable. A further discussion on the resulting system behavior is provided.

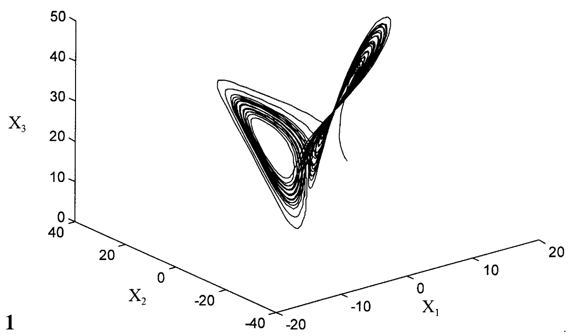
2. Lorenz system

In 1963, Lorenz proposed a simple model for the unpredictable behavior of the weather. He used fluid convection theory to model the motion of a two-dimensional cell of fluid cooled from above and warmed from below. If x_1 represents the convective fluid motion, x_2 denotes the horizontal temperature variation and x_3 denotes the vertical temperature variation, then the non-dimensional forms of Lorenz's equations can be written as

$$\begin{aligned}
\dot{x}_1 &= -\sigma x_1 + \sigma x_2, \\
\dot{x}_2 &= r x_1 - x_2 - x_1 x_3, \\
\dot{x}_3 &= x_1 x_2 - b x_3,
\end{aligned} \tag{1}$$

σ and r are related to the Prandtl number and Rayleigh number, respectively, and the third parameter b is a geometric factor.

A Lorenz system performs complex dynamics, depending on the parameter values, which are often studied for r in the range of $0 < r < \infty$, with σ and b held constant. For $0 < r < 1$, the origin of the system is a stable equilibrium point. For $1 < r < r^*$, the system has one unstable equilibrium point at the origin and two stable equilibrium points at



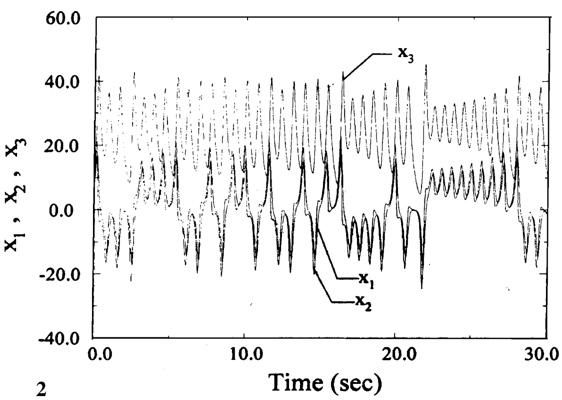


Fig. 1. Chaotic motion of uncontrolled Lorenz system.

Fig. 2. The time series of uncontrolled Lorenz system.

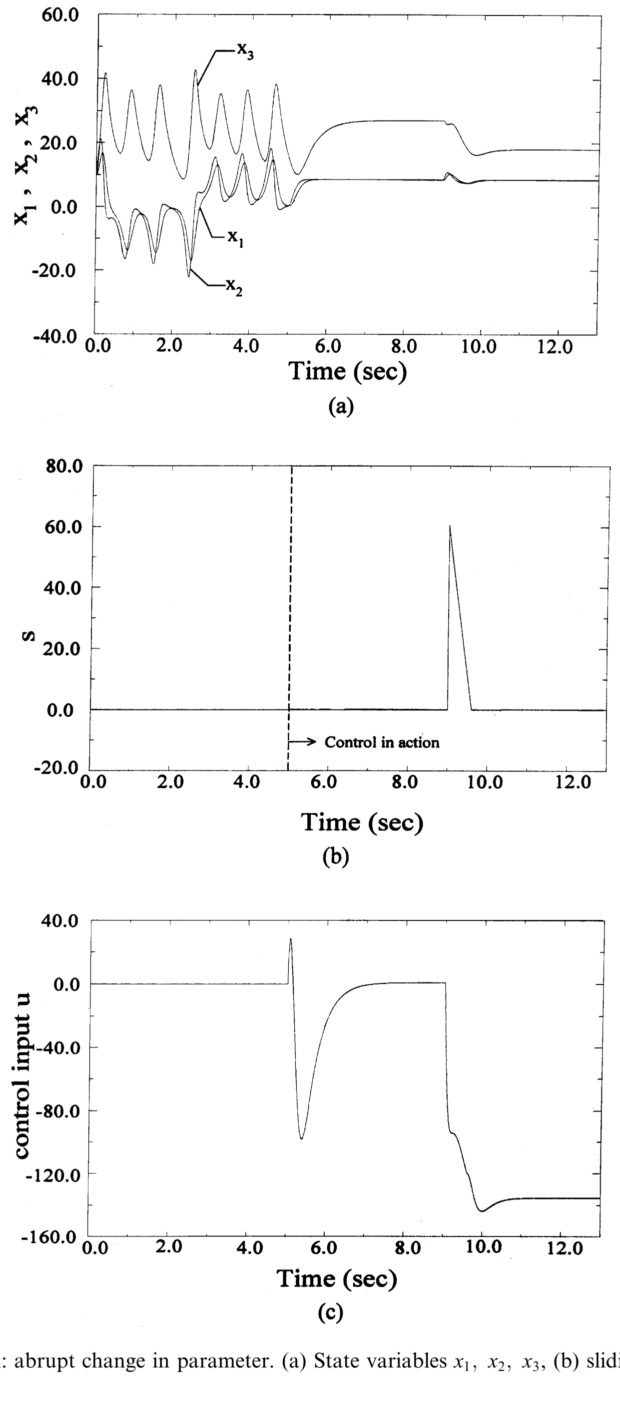


Fig. 3. Sliding mode control: abrupt change in parameter. (a) State variables $x_1,\ x_2,\ x_3$, (b) sliding surface, (c) control input.

$(\pm\sqrt{b(r-1)}, \pm\sqrt{b(r-1)}, r-1)^{\mathrm{T}}$. When $r > r^*$, the other two equilibrium points become unstable spirals and a complex chaotic trajectory forms around all three equilibrium points. For example, if $\sigma = 10$ and $b = \frac{8}{3}$, then

$$r^* = \frac{\sigma(\sigma + b + 3)}{(\sigma - b - 1)} = 24.74.$$

If r is greater than r^*, 24.74 in this case, the dynamic behavior of the Lorenz system will exhibit a two-lobed pattern called the butterfly effect, as is shown in Figs. 1 and 2 with the parameters $\sigma = 10$, $b = \frac{8}{3}$ and $r = 28$.

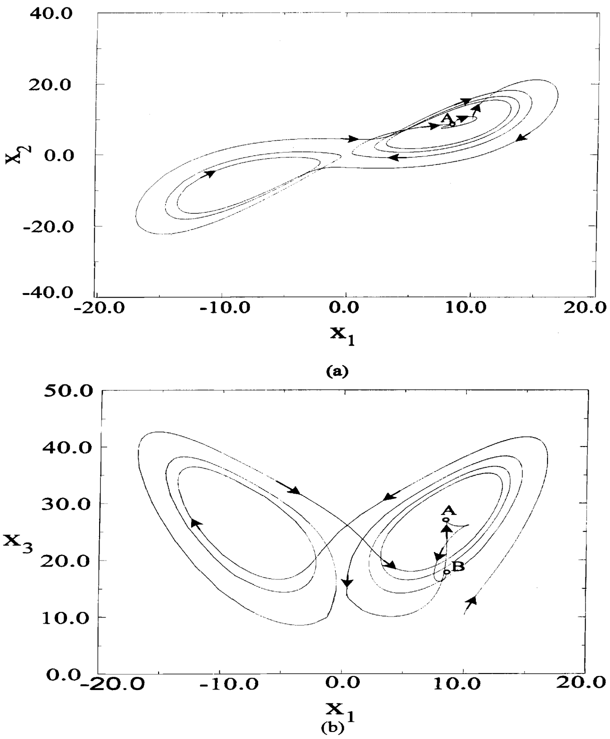


Fig. 4. Sliding mode control: abrupt change in parameter. (a) Phase plot at x_1–x_2 plane, (b) phase plot at x_1–x_3 plane.

3. Control of Lorenz chaos

A Lorenz system with disturbance can be described as

$$\begin{aligned}\dot{x}_1 &= -\sigma x_1 + \sigma x_2 + d_1,\\ \dot{x}_2 &= r x_1 - x_2 - x_1 x_3 + d_2,\\ \dot{x}_3 &= x_1 x_2 - b x_3.\end{aligned} \tag{2}$$

The disturbances d_1 and d_2, mismatched and matched, respectively, are bounded, that is,

$$\begin{aligned}|d_1| &\leqslant \delta_1,\\ |d_2| &\leqslant \delta_2,\end{aligned}$$

where the parameters δ_1 and δ_2 constants. To change the chaotic behavior and regulate the state of the uncontrolled system (Eq. (2)), the proposed method adds a control-input u to the differential equation of state x_2. The controlled system thus becomes

$$\begin{aligned}\dot{x}_1 &= -\sigma x_1 + \sigma x_2 + d_1,\\ \dot{x}_2 &= r x_1 - x_2 - x_1 x_3 + d_2 + u,\\ \dot{x}_3 &= x_1 x_2 - b x_3.\end{aligned} \tag{3}$$

The addition of u will be used to provide a sliding mode controller to drive the system state to a specified point in the state space, even when the system is experiencing match or mismatch disturbance.

It is desired that x_1 be regulated to x_{1r}, where x_{1r} is a given constant. From Eq. (1), if $x_1(t) = x_{1r}$, then $\dot{x}_1(t) = 0$ and $x_2(t) = x_{2r} = x_{1r}$. For $x_1(t) = x_2(t) = x_{1r}$, solving the differential equation for x_3 results in

$$x_3(t) = \mathrm{e}^{-bt} x_3(0) + \frac{x_{1r}^2}{b}(1 - \mathrm{e}^{-bt}),$$

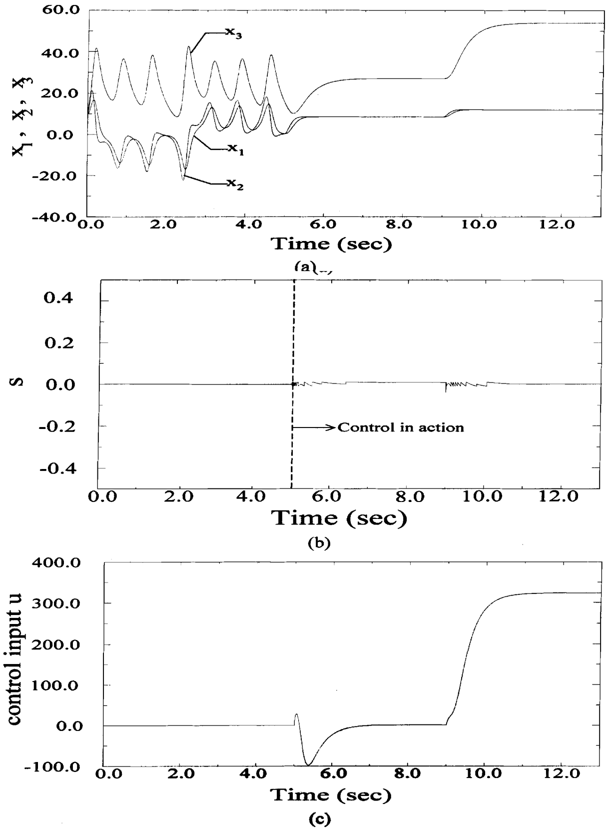


Fig. 5. Sliding mode control: changing set points. (a) State variables $x_1,\ x_2,\ x_3$, (b) sliding surface, (c) control input.

showing that $x_3(t)$ converges to $x_{3r} = x_{1r}^2/b$ when the time $t \to \infty$. Therefore, for a regulated point $x_1(t) = x_{1r}$, the control law seeks to steer the trajectory to the equilibrium point $x_r = (x_{1r}, x_{1r}, x_{3r})^{\mathrm{T}}$, where $(\cdot)^{\mathrm{T}}$ denotes transposition of $(\cdot)$.

3.1. System standardization

To determine the control law, reformulation of the system's original state space equation into an extended controllable canonical form is required. Let the control input $u = u_1 + u_2$ and $u_1 = x_1(t) \cdot x_3(t)$ such that the non-linear term is cancelled, then the dynamic equation with disturbances d_1 and d_2 becomes

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\sigma & \sigma \\ r & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_2, \tag{4}$$

$$\dot{x}_3 = x_1 x_2 - b x_3. \tag{5}$$

From Eq. (4), it can be seen that the dynamics of $x_1(t)$ and $x_2(t)$ are decoupled from $x_3(t)$. Eqs. (4) and (5) show that $x_3(t)$ represents the internal dynamics of the control system, which will converge and stabilize when $x_1(t)$ and $x_2(t)$ converge to x_{1r}.

To get the canonical form, let the state transformation for the system be defined as

$$\mathbf{Z} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \mathbf{P} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \tag{6}$$

where $\mathbf{Z}$ is the new state vector and the transition matrix is

$$\mathbf{P} = \begin{bmatrix} 1/\sigma & -1 \\ 0 & 1 \end{bmatrix}. \tag{7}$$

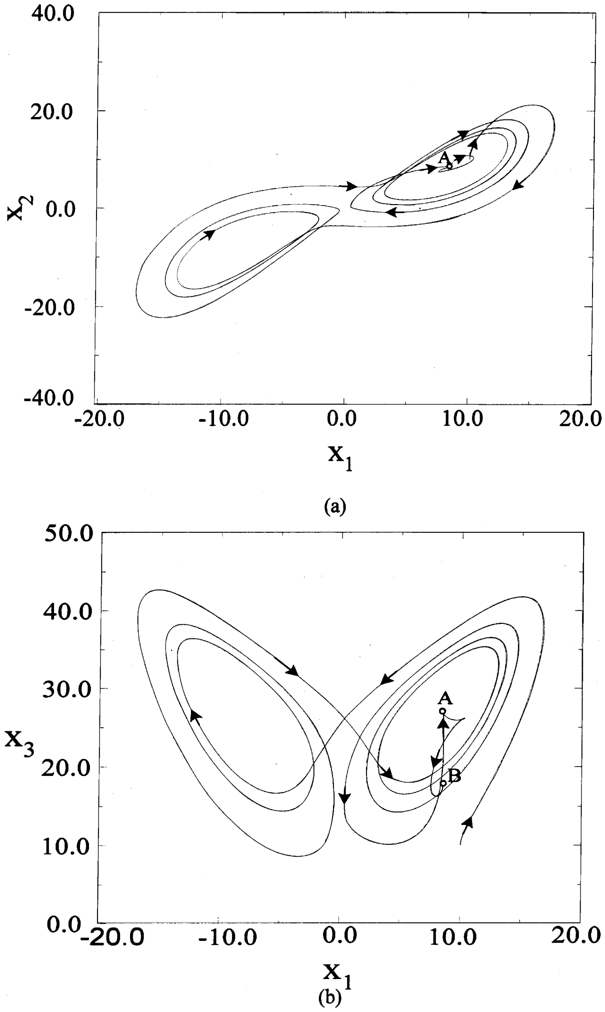


Fig. 6. Sliding mode control: changing set points. (a) Phase plot at x_1–x_2 plane, (b) phase plot at x_1–x_3 plane.

Then the dynamics equation (4) becomes

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \sigma(r-1) & -(\sigma+1) \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \begin{bmatrix} d_1/\sigma \\ -d_1 + d_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_2, \tag{8}$$

a controllable canonical form. If the regulation point is $\mathbf{x}_r = (x_{1r}, x_{1r}, x_{3r})$, then the transformed regulation state vector $\mathbf{Z}_r$ is

$$\begin{aligned} z_{1r} &= \frac{1}{\sigma} x_{1r}, \\ z_{2r} &= -x_{1r} + x_{2r} = 0. \end{aligned} \tag{9}$$

Letting the error states be $e_1 = z_1 - z_{1r}$, $e_2 = z_2 - z_{2r} = z_2$ and $\bar{e} = x_3(t) - x_{3r}$, then the controllable canonical form of the error state dynamic equations is

$$\begin{aligned} \dot{e}_1 &= \dot{z}_1 - \dot{z}_{1r} = z_2 + \frac{d_1}{\sigma} = e_2 + \frac{d_1}{\sigma}, \\ \dot{e}_2 &= \dot{z}_2 = \sigma(r-1)e_1 - (\sigma+1)e_2 + \sigma(r-1)z_{1r} + (-d_1 + d_2) + u_2, \end{aligned} \tag{10}$$

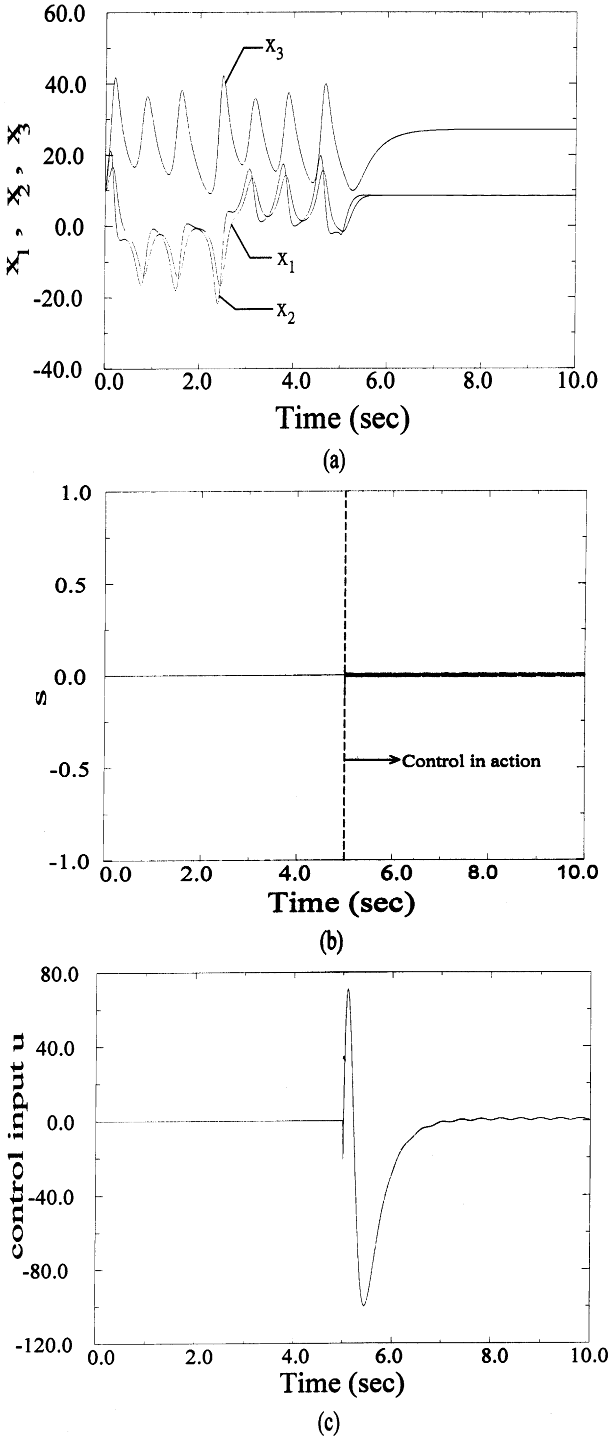
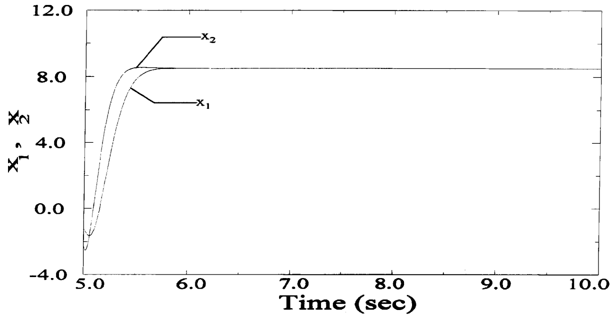
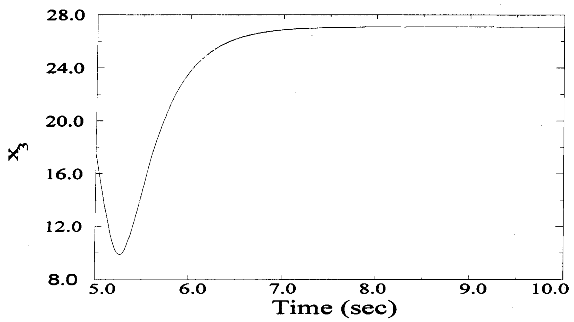


Fig. 7. Sliding mode control: with match disturbance $d_2 = 0.5\cos(5\pi t)$. (a) State variables x_1, x_2, x_3, (b) sliding surface, (c) control input.



(a)



(b)

Fig. 8. Sliding mode control: time response at $t \in [5, 10]$ with match distrubance $d_2 = 0.5\cos(5\pi t)$. (a) State variables $x_1,\ x_2$, (b) state variable x_3.

and the internal dynamics of the error state is

$$\begin{aligned}\dot{\bar{e}} &= -b\bar{e} - bx_{3r} + \sigma e_1 e_2 + \sigma e_2 z_{1r} + \sigma^2 (e_1 + z_{1r})^2 \\ &= -b\bar{e} + g(e_1, e_2),\end{aligned} \tag{11}$$

where

$$g(e_1, e_2) = -bx_{3r} + \sigma e_1 e_2 + \sigma e_2 z_{1r} + \sigma^2 (e_1 + z_{1r})^2.$$

Letting $\dot{e}_2 = e_3$, then standardized state space equations of the error state are:

$$\begin{aligned}\dot{e}_1 &= e_2 + \frac{d_1}{\sigma}, \\ \dot{e}_2 &= e_3,\end{aligned} \tag{12}$$

$$\begin{aligned}\dot{e}_3 &= \sigma(r-1)e_2 - (\sigma+1)e_3 + (-\dot{d}_1 + \dot{d}_2) + \dot{u}_2, \\ \dot{\bar{e}} &= -b\bar{e} + g(e_1, e_2).\end{aligned} \tag{13}$$

where Eq. (12) represents an extended controllable canonical form and Eq. (13) is the system internal dynamics.

3.2. Sliding surface design and associated control law

Based on the control law and using the concept of extended systems [11], a sliding surface suitable for this application can be defined as

$$s = e_3 - e_3(0) + \int_0^t (c_3 e_3 + c_2 e_2 + c_1 e_1)\,\mathrm{d}t. \tag{14}$$

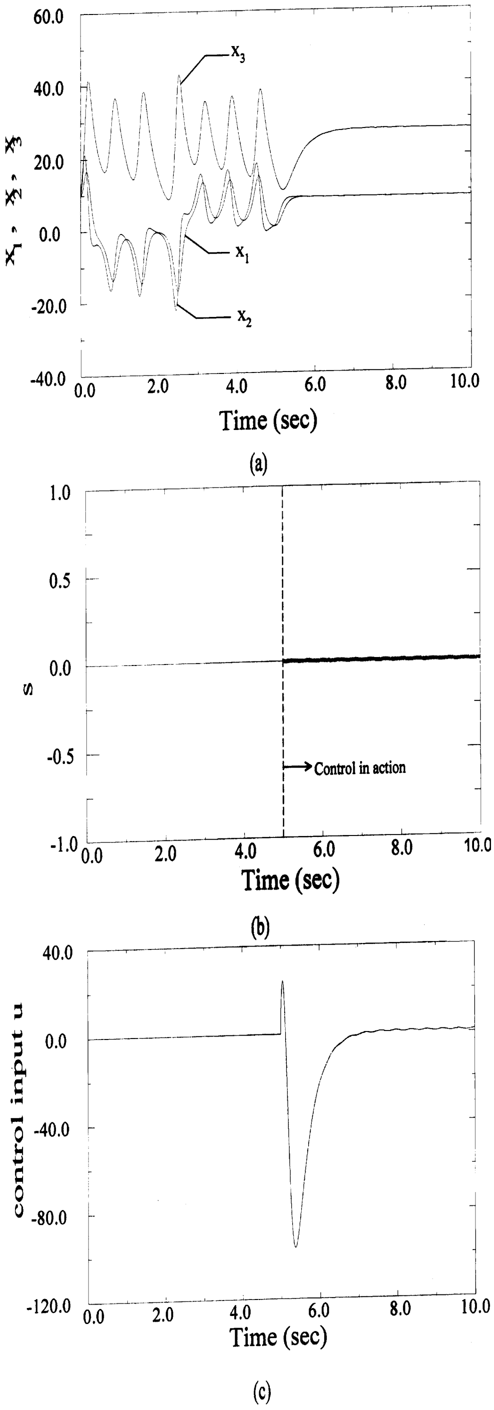


Fig. 9. Sliding mode control: with mismatch distrubance $d_1 = 0.5\cos(5\pi t)$. (a) State variables $x_1,\ x_2,\ x_3$, (b) sliding surface, (c) control input.

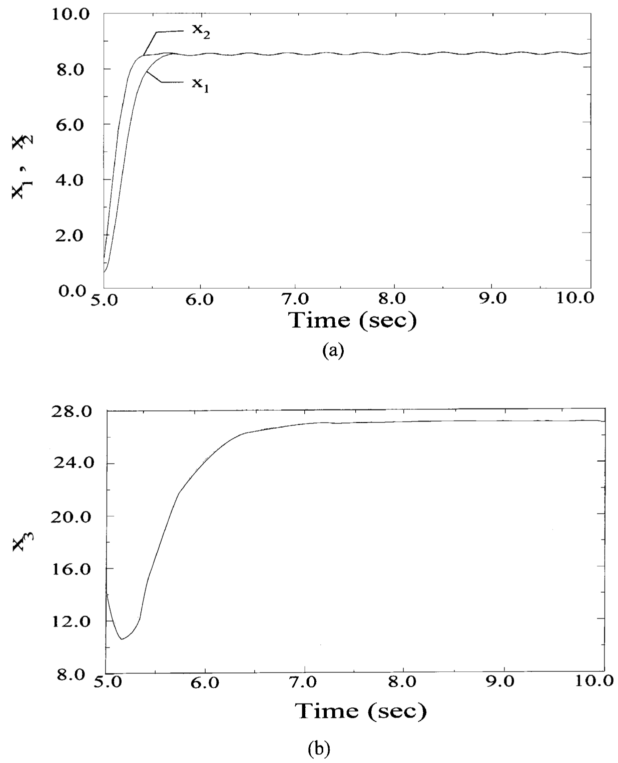


Fig. 10. Sliding mode control: time response at $t \in [5, 10]$ with mismatch distrubance $d_1 = 0.5\cos(5\pi t)$. (a) State variables x_1, x_2, (b) state variable x_3.

The reaching law can be chosen as

$$\dot{s} = -w\,\mathrm{sgn}(s). \tag{15}$$

From Eqs. (14) and (15), it can be found that

$$\begin{aligned}\dot{s} &= -w\,\mathrm{sgn}(s)\\ &= \sigma(r-1)e_2 - (\sigma+1)e_3 + (-\dot{d}_1 + \dot{d}_2) + \dot{u}_2 + c_3e_3 + c_2e_2 + c_1e_1,\end{aligned} \tag{16}$$

and $\dot{u}_2$ can be obtained as

$$\dot{u}_2 = -\sigma(r-1)e_2 + (\sigma+1)e_3 - (-\dot{d}_1 + \dot{d}_2) - (c_3e_3 + c_2e_2 + c_1e_1) - w\,\mathrm{sgn}(s)$$

with the initial condition $u_2(0) = 0$. In real application, disturbances d_1 and d_2 are unknown and the implemented control input is described by

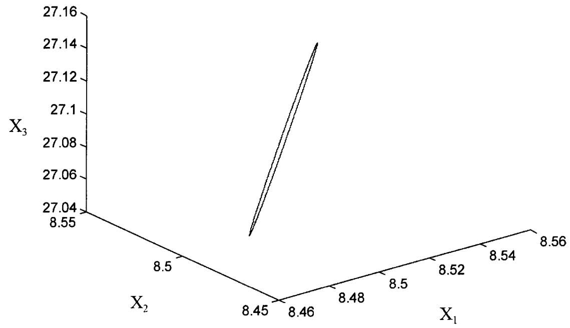
$$\dot{u}_2 = -\sigma(r-1)e_2 + (\sigma+1)e_3 - (c_3e_3 + c_2e_2 + c_1e_1) - w\,\mathrm{sgn}(s). \tag{17}$$

That is

$$\begin{aligned}u_2 &= \int_0^t [-\sigma(r-1)e_2 + (\sigma+1)e_3 - w\,\mathrm{sgn}(s) - (c_3e_3 + c_2e_2 + c_1e_1)]\,\mathrm{d}t,\\ u_2(0) &= 0.\end{aligned} \tag{18}$$

Therefore,

$$\begin{aligned}u &= u_1 + u_2\\ &= x_1 \cdot x_3 + \int_0^t [-\sigma(r-1)e_2 + (\sigma+1)e_3 - w\,\mathrm{sgn}(s) - (c_3e_3 + c_2e_2 + c_1e_1)]\,\mathrm{d}t.\end{aligned} \tag{19}$$



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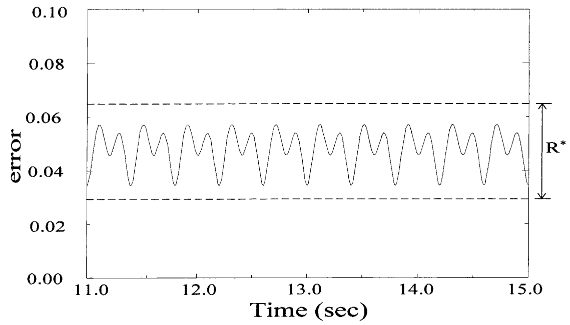

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Fig. 11. The phase plane trajectory at $t \in [11, 15]$ with mismatch distrubance $d_1 = 0.5\cos(5\pi t)$.
Fig. 12. The error bound of system vector relates to equilibrium point with mismatch distrubance $d_1 = 0.5\cos(5\pi t)$.

3.3. Stability analysis

Substituting the control law (17) into the extended system (12), the closed-loop system dynamics can be described as

$$\begin{aligned} \dot{e}_1 &= e_2 + \frac{d_1}{\sigma}, \\ \dot{e}_2 &= e_3, \\ \dot{e}_3 &= -w\,\mathrm{sgn}(s) - (c_3e_3 + c_2e_2 + c_1e_1) + (-\dot{d}_1 + \dot{d}_2). \end{aligned} \tag{20}$$

Letting the Lyapunov function of the system be $V = \frac{1}{2}s^2$, then its first derivative with respect to time is

$$\begin{aligned} \dot{V} &= s\dot{s} = s(-w\,\mathrm{sgn}(s)) \\ &= s(\dot{e}_3 + c_3e_3 + c_2e_2 + c_1e_1) \\ &= s(-\dot{d}_1 + \dot{d}_2 - w\,\mathrm{sgn}(s)) \\ &= -w|s| + s(\dot{d}_2 - \dot{d}_1) \\ &\leqslant |s|(|\dot{d}_2| + |\dot{d}_1| - w). \end{aligned} \tag{21}$$

Therefore, if

$$w \geqslant |\dot{d}_1| + |\dot{d}_2|, \tag{22}$$

then $\dot{V} \leqslant 0$, confirming the presence of sliding mode dynamics. Since $\dot{V}$ is negative semi-definite, it follows that the error states $e_1(t)$ and $e_2(t)$ are bounded. It may further be concluded that there exists a function *h(t)* such that $g(e_1, e_2)$ in Eq. (11) satisfies

$$|g(t)| \stackrel{\Delta}{=} |g(e_1, e_2)| \leqslant h(t). \tag{23}$$

From Eq. (13), it can be concluded that the error state of internal dynamics, $\bar{e}(t)$ will be bounded when $g(e_1, e_2)$ is bounded.

4. Simulation results

In this section, simulation results are presented. For the uncontrolled Lorenz system, the parameters are $\sigma = 10$, $b = \frac{8}{3}$, and $r = 28$. Let $w = 10$ and the eigenvalues corresponding to the sliding surface be -10, $-10 - 6i$, $-10 + 6i$, which results in $[c_3,\ c_2,\ c_1] = [30,\ 336,\ 1360]$.

4.1. Regulation for different values of (σ, r, b)

Combining Eqs. (3) and (19) gives the simulated closed-loop system. Initially, disturbance is set at $d_1 = 0$, $d_2 = 0$. For the first five seconds (Fig. 3), the system is uncontrolled and the trajectories are chaotic. The control input becomes

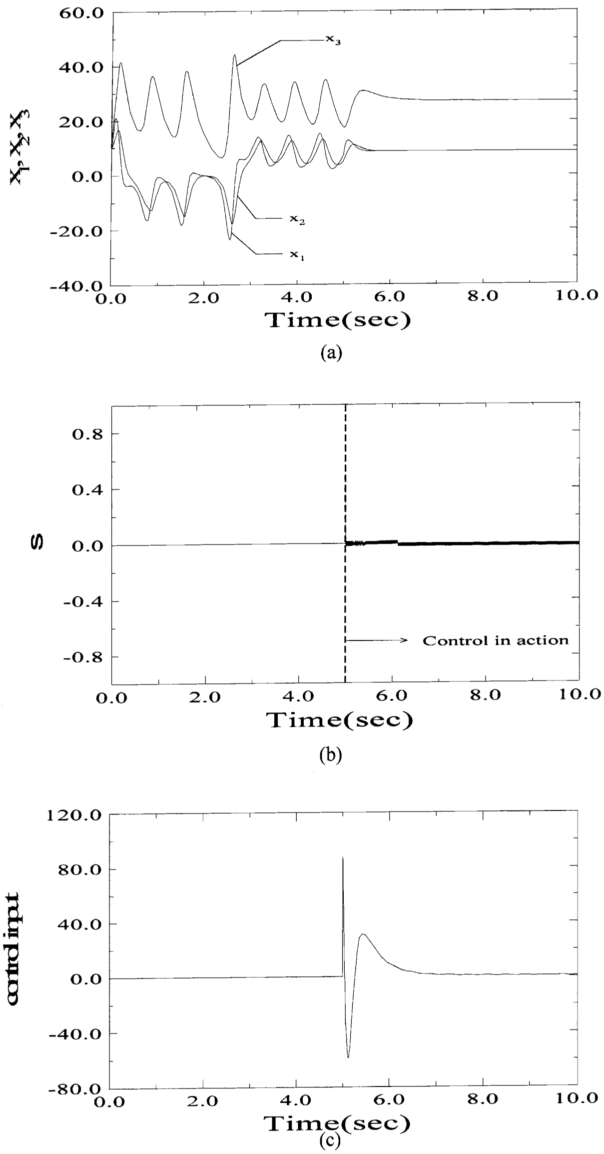

Fig. 13. Sliding mode control: with mismatch distrubance $d_3 = 0.5\cos(5\pi t)$. (a) State variables x_1, x_2, x_3, (b) sliding surface, (c) control input.

active at $t = 5$ s, with the objective of regulating the system state to $x_r = (x_{1r}, x_{1r}, x_{1r}^2/b)^{\mathrm{T}}$. The initial parametric values are $\sigma = 10,\ b = \frac{8}{3}$ and $r = 28$. These parameters abruptly change to $\sigma = 15,\ b = \frac{12}{3}$ and $r = 35$ at $t = 9$ s. Fig. 3(a) shows that the chaotic behavior of the uncontrolled system disappears at $t = 5$ s. The state is first regulated to $x_r = (8.5, 8.5, 27.1)^{\mathrm{T}}$ and then, following the change in parameters, to the new desired state $x_r = (8.5, 8.5, 18.1)^{\mathrm{T}}$. The abrupt change in system parameters at $t = 9$ s causes only a small transient response, and the state vector converges smoothly to the desired state. The duration of the transient response can be controlled by assigning the corresponding eigenvalues of the sliding surfaces. Transient duration is shorter than 1 s in this case. Fig. 3(a) also shows that x_1 and x_2 converge to the same value, but x_3 converges to a new b-dependent value. The sliding surface dynamics are depicted in Fig. 3(b), showing that the change in parameters at $t = 9$ s causes a dramatic change in the sliding surface dynamics, decaying rapidly to 0. The corresponding control input is continuous, as described in Fig. 3(c). It also shows that chattering does not appear, due to the continuous control. The phase plane trajectory is shown in Fig. 4. It can be seen that the system state vector initially stabilizes at point A. The parameter change at $t = 9$ s results in the trajectory in the

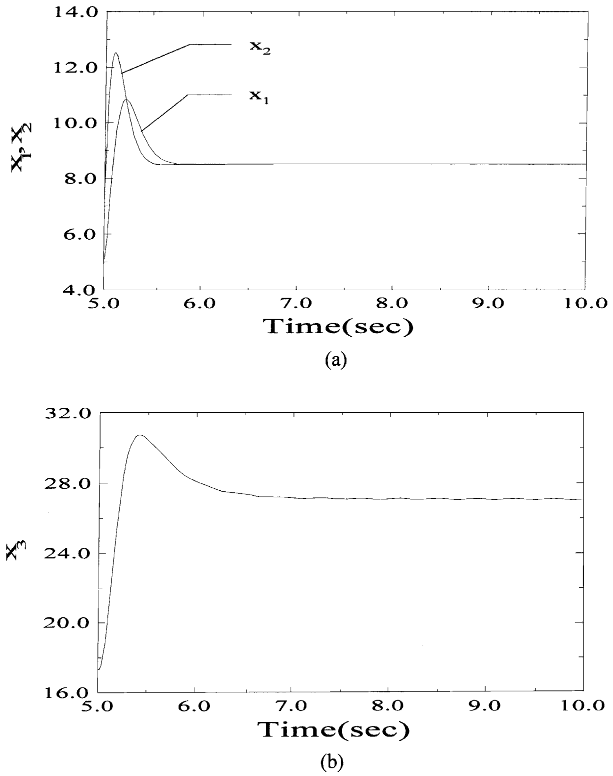

Fig. 14. Sliding mode control: time response at $t \in [5, 10]$ with mismatch distrubance $d_3 = 0.5\cos(5\pi t)$. (a) State variables x_1, x_2, (b) state variable x_3.

x_1–x_2 plane leaving point A during the transient response and returning again to point A when the transient is over. However, the trajectory establishes a new equilibrium point, B, in the x_1–x_3 plane.

4.2. Regulation to set points

In the second simulation, for fixed parameter values $\sigma = 10$, $b = \frac{8}{3}$ and $r = 28$ without disturbance, it is desired to steer the system state to a set point ($x_r = (8.5, 8.5, 27.1)^{\mathrm{T}}$) and then to move it to a new set point ($x_r = (12, 12, 54)^{\mathrm{T}}$). Fig. 5(a) shows that the control force becomes active at $t = 5$ s, with x_1 and x_2 both stabilizing to 8.5 within one second of control onset, and the internal state x_3 converging to 27.1 in the same time interval. At $t = 9$ s, a new reference value $x_{1r} = 12$ becomes active and, within one second, x_1 converges smoothly to 12. The sliding surface and control input are shown in Fig. 5(b) and (c). The phase plots are shown in Fig. 6. It can be seen that the system state initially stabilizes at point A and then, following the new reference command, stabilizes at point B.

4.3. Regulation with disturbance

In the third simulation, a closed-loop system with disturbance is considered. Fig. 7(a)–(c) show the response of the closed-loop system with match disturbance $d_2 = 0.5\cos(5\pi t)$ and mismatch disturbance $d_1 = 0$. It can be seen that the corresponding control input has a fixed non-zero demand in the steady state due to the sinusoid disturbance. It also shows that chattering does not appear with this continuous control. The interval $t = 5$–10 s is shown in Fig. 8, showing that the system state is successfully regulated to the equilibrium point $x_r = (8.5, 8.5, 27.1)^{\mathrm{T}}$, even when the system is undergoing match disturbance.

Fig. 9 shows the response of the closed-loop system with mismatch disturbance $d_1 = 0.5\cos(5\pi t)$ and match disturbance $d_2 = 0$ (see also Fig. 13). It also shows a harmonic waveform for the control input in the steady state. Due to the mismatch disturbance, the steady state of the system state vector manifests a trajectory very close to the desired equilibrium point, but does not stabilize at it. This steady-state vibration in the time interval $t = 5$–10 s is clearly shown in Fig. 10(a), and is in sharp contrast with Fig. 8(a)'s constant value (see also Fig. 14). With mismatch disturbance, the steady-state phase plane trajectory describes a limit cycle, as shown in Fig. 11. The system error state norm to the equilibrium point is shown in Fig. 12 and can be seen to be bounded by radius R^*.

5. Conclusion

In this paper, a sliding mode control system for Lorenz chaos is proposed. Based on the Lyapunov stability theory, a sliding mode controller is designed for the regulation of the state vector to a desired point in the state space. Simulation results show that the proposed controller is able to stabilize Lorenz chaos, even when the system is undergoing match or mismatch disturbance. The chattering phenomenon of conventional switching type sliding controls does not occur in this study. Under the proposed control method, the settling time of regulation can be arbitrarily determined by assigning the corresponding eigenvalues of the sliding surfaces. However, a suitable sliding surface should be chosen such that the control input will not exceed the system's physical limitations.

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